

INSTITUTE OF ACTUARIES OF INDIA

EXAMINATIONS

May 2026

SA4- Pensions and Other Benefits

Time allowed: 3 Hours 15 Minutes

Total Marks: 100

Q.1) The table given below shows the life expectancy of a country's population compared over the years 1950 & 2000.

Life Expectancy of population (years)		
age	in the year 1950	in the year 2000
New born	65.5	74.4
25	44.4	50.9
35	35.2	41.5
45	26.6	32.5
55	19	24
65	12.7	16.4
75	7.8	10.2
85	4.4	5.7
95	0	2.5
>100	0	1.6

- i) Comment on the trends in the longevity with illustrations. [3]
- ii) Discuss how increasing trends in longevity are likely to reshape the pensions as a retirement benefit, in future. [12]

A global company "X" having business across different global regions including India is having pension scheme for its employees administered through Trustees. It offers Defined benefit & Defined contribution pension benefits depending upon the industry practices prevailing in each region. The trust is using commutation tables prevailing in each region to pay the commutation benefits. The tables have not been revised for the last 20 years.

Now there is a demand from members of the scheme to revise the commutation factors to allow for the trends in longevity. As a consultant actuary you have been asked to examine the scope for updating the commutation factors.

- iii)
- a) Discuss the step by step process to be taken for revising the commutation factors. [7]
- b) The Trustees are also examining the feasibility of adopting a single table across all region/ to pay the commutation benefit for the company. Discuss the proposal of the Trustees & the factors to be considered by Trustees before adopting a single table [6]

In India, New Labour codes have been introduced; Management of the company X wants to understand the implications to the company. You as a consultant actuary have been invited to give a presentation on the features of the new laws.

- iv) State the points to be included in the presentation highlighting the impact on short term & long term employee benefits [8]

Company X is now planning to exit from Indian markets by selling its Indian subsidiary (Ind X). The company proposes to retain its accrued pension liabilities under the existing Trustee arrangements. However, Trustees are reluctant to the proposals of company X to administer the

pension schemes on “closed fund” basis. A brief profile & status of the pension scheme of the Indian subsidiary (Ind X) as on 31.12.2025 are given as under:

Liability Profile of Ind X (NRA 60 years)						Assets	
Member group	type of benefit provided	Number of members	Average age	Past service	Value of Liabilities (in million)	Nature of assets	market value of assets(in million)
Active (category I)	50% final pay with DA increases;	75	54	25	25	Equities	15
Active (category II)	50% of pay- No pension increase	500	48	20	30	Corporate bonds	25
Deferred guarantee pensioners	Assured pensions	500	40		18	Cash accumulation scheme of insurers	30
Deferred pension on DC basis	DC basis				12	Government bonds	20
Discretionary pensions to pensioners	nominal		75		15	Cash & Short term investments	20
Total liabilities					100	Total assets	110

(You may assume Trustee of Ind X is buying annuities from the insurance companies to provide pension benefits as & when they fall due.)

v)

a) Discuss the possible reasons for the company X to retain the pension liabilities in the sale process & why the Trustees are reluctant of managing the pension fund closed to further accrual. [10]

b) List the other ways how the Company X can deal with the pension benefits of Ind X. [4]
[50 marks]

Q. 2) ABC Ltd, a company operating in Country X, provides lump sum benefits to its employees under a pre-defined formula linked to years of service as given below.

Service of an employee at the time of exit	Benefits
Between 0 years and 5 years	3 months' Salary
Between 5 years and 10 years	9 months' Salary
Between 10 years and 20 years	15 months' Salary
More than 20 years	20 months' Salary

- i) State the relevant provisions of IAS 19 – Employee Benefits concerning the attribution of benefits to periods of service & Explain how the benefit is to be attributed for an employee joining at age 25 & expected to be with company till retirement age 60 while determining the Present Value of the Defined Benefit Obligation (DBO), and the Current Service Cost (CSC). [6]

The valuation results of lump sum benefits of ABC Ltd., for the year ending 31.12.2024 & 31.12.2025 are given as under:

Particulars	31.12.2024	31.12.2025
Discount Rate	5%	5.50%
Salary Escalation Rate	7%	7.50%
Mortality Table	IALM 2006-08 Table	IALM 2006-08 Table
Attrition Rate	8% for all ages till 59	6% for all ages till 59
PV DBO	300 million USD	525 million USD
Current Service Cost	17 million USD (estimated as at 01.01.2024)	27 million USD (estimated as at 01.01.2025)
Benefit Paid	0	22 million USD
Contribution paid	No Contribution	250 million USD paid on 31.12.2025
Increase in Liability as at 30.06.2025 due to transfer of employees from a sister company.	NIL	120 million USD
Current Service Cost for the transferred employees		14 million USD (Estimated as at 01.01.2025)
Expected average future working life time	12 years	15 years

- ii) Prepare a reconciliation for the change in liabilities (DBO) & asset (if any) for the year ending 31.12.2025 using the above figures. [5]

In December 2025, a change in government policy led ABC Ltd to amend its benefits & the rules effective from 31 December 2025. As a result, a revised actuarial valuation was carried out as per IAS 19. The amendment generated a **Past Service Cost** of USD 150 million, and the (recalculated) **Defined Benefit Obligation (DBO)** is USD 675 million as at 31 December 2025. Due to this revision the company wants to understand the implications of the change in accounting for the same period prepared both under IAS 19 & under ASC 715 – Compensation – Retirement Benefits (US GAAP).

- iii) Compare the following components in terms of IAS 19 – Employee Benefits and ASC 715 – Compensation – Retirement Benefits with respect to the recognition and presentation:
- Past Service Cost,
 - Actuarial Gains and Losses,
 - the Statement of Profit or Loss, and
 - Other Comprehensive Income (OCI),

and calculate the above components for the same period under both IAS 19 and ASC 715 using the above table. [20]

The Finance Director of the company suggests an annual contribution amount based on the current service cost as measured every year and an additional contribution based on amortized cost (of past service cost and the actuarial loss) will be sufficient to meet the liabilities as & when the benefit falls due in the next 15 years.

- iv) Discuss the validity of the statement and the factors impacting the funding of the scheme. [10]

ABC Ltd's Indian subsidiary operates a stock option scheme for its senior management officials. Under the scheme, each eligible official who has completed three years of service is entitled to 100 stock options.

- v) State the key points of the Guidance notes issued by the Institute of Actuaries of India to the actuaries while developing basis for valuation of fair value of stock option (as per Ind AS 102) provided by the company to its employees. [9]

[50 marks]
